1. Suppose M be the set of all possible plaintexts, K the set of keys, and C the set of ciphertexts in a cryptographic system.

(i) State Shannon's definition of perfect secrecy mathematically.

(ii) Prove that the one-time pad achieves perfect secrecy under the condition that the key K is uniformly random and

$$|K| \geq |M|$$

critically analyze why perfect secrecy is impractical for large-scale communication systems.

<u>Ans:</u>

Let, M = plaintext space

K = key space

C = ciphertext space

(i) A cryptosystem has perfect secrecy if

$$P(M=m \mid C=c) = P(M=m)$$

for all $m \in M$ and $c \in C$

Equivalent form:

$$P(C=c \mid M=m_1) = P(C=c \mid M=m_2)$$

for all m_1, m_2.

(ii) <u>One-Time Pad achieves Perfect secrecy</u>

<u>Encryption:</u> $c = m \oplus k$

$|K| \geq |M|$

key K is uniformly random

for any m, c:

$$P(C = c \mid M = m) = P(K = c \oplus m)$$

since K uniform:

$$P(K = k) = \frac{1}{|K|}$$

Thus ciphertext distribution independent of plaintext.

Therefore:

$$P(M = m \mid C = c) = P(M = m)$$

so, OTP provides perfect secrecy.

(iii) Why Perfect secrecy is Impractical:

- Key length $\geq$ message length
- Keys must be truly random.
- Keys used only once.
- secure key distribution difficult
- Large storage requirement.

Hence impractical for large system.

2. A linear congruential Generator (LCG) is defined by the recurrance relation: $x_{n+1} = ax_n + c \bmod m$ where x_n is the sequence of Pseudo random numbers, and a, c and m are integer Parameters representing the multiplier, increment, and modulus, respectively. Using specific values for aaa, ccc, and mmm, compute the first 5 numbers of an LCG sequence starting with a given seed $x_0 = 7$

Ans:

formula: $x_{n+1} = a x_n + c \bmod m$

Given typical values:

$a = 5$, $c = 3$, $m = 16$, $x_0 = 7$

Compute:

$$x_{n+1} = ax_n + c \bmod m$$

$$x_1 = (5(7) + 3) \bmod 16$$

$$= 38 \bmod 16 = 6$$

$$x_2 = (5(6) + 3) \bmod 16$$

$$= 33 \bmod 16$$

$$= 1$$

$$x_3 = (5(1) + 3) \bmod 16$$

$$= 8 \bmod 16 = 8$$

$$x_4 = (5(8) \cancel{\bmod} + 3) \bmod 16$$

$$= 43 \bmod 16 = 11$$

$$x_5 = (5(11) + 3) \bmod 16$$

$$= 58 \bmod 16 = 10$$

sequence: 6, 1, 8, 11, 10

3. Suppose G be a group, and let H be a subgroup of G. Prove that the intersection of any two subgroups of G is also a subgroup of G. Provide an example using specific groups.

Ans:

Let H_1, H_2 be subgroups of G.

~~Cons~~ consider: $H = H_1 \cap H_2$

Proof:

1. $e \in H_1$ and $e \in H_2$

Thus $e \in H$

2. closure:

If $a, b \in H$

Then $a, b \in H_1$ and H_2

So, $ab^{-1} \in H_1$

and $ab^{-1} \in H_2$

Thus $ab^{-1} \in H$

Therefore H is a subgroup.

Example:

Let $G = \mathbb{Z}$

$H_1 = 3\mathbb{Z}$

$H_2 = 2\mathbb{Z}$

Then $H_1 \cap H_2 = 6\mathbb{Z}$

which is also a subgroup

4. The ring $\mathbb{Z}_n$ is commutative and identify whether it has zero divisors, Prove. Further, determine the conditions under which $\mathbb{Z}_n$ is a field.

Ans:

The ring $\mathbb{Z}_n$ is commutative:

$$a+b = b+a$$

$$ab = ba$$

Zero Divisors:

If n composite

Example in $\mathbb{Z}_6$:

$$2 \times 3 = 0 \bmod 6$$

So zero divisors exist.

Z_n is field iff n is prime.

Example: Z_5

All elements have multiplicative inverse.

5. consider the group S_4 acting on the set of all subsets of $\{1,2,3,4\}$ by permutation. Prove that this action is well-defined and compute the orbit and stabilizer of the subset $\{1,2\}$

Ans:

$$\sigma(A) = \{\sigma(a) : a \in A\}$$

for subset $A \subseteq \{1,2,3,4\}$

orbit of $\{1,2\}$

All 2-element subsets:

$\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}$

orbit size = 6

Stabilizer

permutations that keep set $\{1,2\}$ unchanged.

Possible maps inside set.

Elements: $(1), (1\ 2), (3\ 4), (1\ 2)(3\ 4)$

size = 4

6. Let $GF(2^2)$ be the finite field of order 4, constructed using the irreducible polynomial x^2+x+1 over $GF(2)$

(i) Show that $GF(2^2)$ forms a group under multiplication

(ii) Verify whether the set of all nonzero elements of $GF(2^2)$ is cyclic.

<u>Ans:</u> construct using : x^2+x+1

Elements : $\{0, 1, x, x+1\}$

(i) <u>Group under multiplication</u> :

Non-zero elements : $\{1, x, x+1\}$

closed under multiplication $\rightarrow$ group.

(ii) It is cyclic

Generator : x

since x^1

$x^2 = x+1$

$x^3 = 1$

Thus cyclic group.

7. Let $GL(2,R)$ be the general linear group of 2×2 invertible matrices over R. Show that the set of scalar matrices forms a normal subgroup of $GL(2,R)$. Construct the corresponding factor group, and interpret its structure.

Ans:

Scalar matrices:

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

Let $A \in GL(2,R)$

$$A(\lambda I)A^{-1} = \lambda A I A^{-1} = \lambda I$$

Hence Invariant

Thus scalar matrices form normal subgroup.

CamScanner

8. Consider the following modes of operation for block ciphers: ECB (Electronic codebook) CBC (Cipher Block Chaining) and CTR (counter mode)

(i) For a block cipher with block size n, mathematically prove why ECB mode is insecure for encrypting highly redundant data.

Ans:

Encryption:

$$C_i = E_K(P_i)$$

Same plaintext blocks produce same ciphertext.

Thus patterns preserved.

Example: image encryption still visible.

Therefore insecure for redundant data.

Therefore insecure for redundant data.

CamScanner

(ii) Derive the recurrence relation for CBC mode encryption and decryption and prove that error propagation is limited in decryption.

Ans:

Encryption : $C_i = E_K (P_i \oplus C_{i-1})$

with $C_0 = IV$

Decryption: $P_i = D_K (C_i) \oplus C_{i-1}$

Error Propagation

Error in C_i : → corrupts P_i
→ Partially affects P_{i+1}

After that blocks uneffected.

CamScanner

q. Given an RSA key pair with public key (e,n) and the private key (d,n), where: $p = 5$, $q = 11$ (two prime numbers), $n = p \cdot q$ and $\phi(n) = (p-1)(q-1)$. use the RSA algorithm to encrypt a message, $m = 2$ and decrypt the ciphertext to recover the original message.

Ans: Given, $p = 5$, $q = 11$

$$n = pq = 5 \times 11 = 55$$

$$\phi(n) = (p-1)(q-1)$$

$$= (5-1) \times (11-1) = 4 \times 10 = 40$$

choose $e = 3$

$$3d \equiv 1 \bmod \phi(n)$$

$$3d \equiv 1 \bmod 40$$

$$d \equiv 27$$

Encrypt: $m = 2$

$$c = 2^e \bmod n$$

$$= 2^3 \bmod 55$$

$$= 8$$

Decrypt

$$m = e^{d} \bmod n$$

$$= 8^{27} \bmod 55$$

$$m = 2$$

10. Let $p = 7$, $q = 3$ sign the hash of a message $H(m) = 3$ using the private key d. verify the signature using the public key (e,n). Explain how the signature ensures the integrity and authenticity of the message.

Given, $p = 7$, $q = 3$

$$n = pq = 7 \times 3 = 21$$

$$\varphi = (p-1)(q-1) = 6 \times 2 = 12$$

Let $e = 5$,

compute d

$$5d = 1 \bmod \varphi$$

$$\Rightarrow 5d = 1 \bmod 12$$

$$\Rightarrow d = 5$$

Hash:

$$H(m) = 3$$

Signature

$$s = 3^{d} \bmod n$$

$$= 3^{5} \bmod 21$$

$$= 243 \bmod 21$$

$$= 12$$

Verification

$$s^{d} \bmod n$$

$$= 12^{5} \bmod 21$$

$$= 3$$

matches hash.

Thus valid.

11. Describe the fundamental security properties of cryptographic hash functions used in the secure Hash Algorithm family, such as SHA-256. In your explanation, define key concepts including Pre-Image resistance, second-pre-image resistance, and collision resistance.

Ans:

(i) Pre-image resistance

Given h, hard to find m

$$H(m) = h$$

(ii) Second Pre-image resistance

Given m_1

Find m_2 $\quad H(m_1) = H(m_2)$

(iii) collision resistance

Hard to find $m_1 \neq m_2$

such that $H(m_1) = H(m_2)$

Example: SHA-256.

CamScanner

12. Analyze how the output size of a hash function (for example, the 256-bit output of SHA-256) influences its resistance to brute-force and collision attacks.

Ans:

For hash length n :-

Preimage attack : 2^n

Collision attack : $2^{n/2}$

For 256-bit hash :

Preimage : 2^{256}

Collision : $2^{256/2} = 2^{128}$

Hence extremely secure.

13. Explain the practical applications of SHA algorithms in modern systems, particularly in digital signature schemes and blockchain technologies.

Ans: common uses of SHA-256:

(i) Digital signatures

(ii) TLS certificates.

(iii) Password Hashing

(iv) File integrity

(v) Block chain

Example: used in Bitcoin mining.

14. Define the shortest vector problem (SVP) and explain why its computational hardness underpins the security of lattice-based cryptographic constructions.

Ans:

In a lattice L find non-zero vector v such that

$$||v||$$

is minimal.

This is the shortest vector problem (SVP)

SVP is NP-hard.

Security of lattice cryptography relies on difficulty of SVP.

15. Compare the underlying security assumptions of lattice-based cryptography with classical public-key schemes such as RSA and Elliptic curve cryptography, particularly in light of quantum attacks enabled by Shor's algorithm.

Ans:

Feature	RSA/ECC	Lattice
Hard Problem	Factoring / Discrete Log	SVP / LWE
Quantum Resistant	NO	Yes
Key size	small	Larger
Speed	moderate	Fast

Quantum algorithm

Shor's algorithm breaks RSA/ECC

but Lattice scheme remains secure.

16. Differentiate between quantum cryptography and lattice-based cryptography, focusing on their objectives, theoretical foundations, and implementation approaches.

Ans:

Quantum cryptography

- uses quantum physics
- Example: Quantum key distribution
- Feature:
  - detects eavesdropping
  - requires quantum hardware

Lattice cryptography

- based on lattice problems.
- Example:
  - Learning with Errors
  - Shortest vector problem.
- Advantages:
  - quantum resistant
  - implementable on classical computers.